

# Contents

|                                             |          |
|---------------------------------------------|----------|
| <b>Counterfactual Reasoning</b>             | <b>1</b> |
| Introduction . . . . .                      | 1        |
| Building the Atlas . . . . .                | 1        |
| Baseline: The Causal Backbone . . . . .     | 2        |
| Hard Do-Cut: Removing Vaccination . . . . . | 3        |
| Do-Cut Diff: What Changes? . . . . .        | 3        |
| Soft Do: Attenuated Intervention . . . . .  | 4        |
| Comparing Multiple Interventions . . . . .  | 4        |
| Interpreting Counterfactuals . . . . .      | 5        |
| Summary . . . . .                           | 5        |

## Counterfactual Reasoning

- Introduction
- Building the Atlas
- Baseline: The Causal Backbone
- Hard Do-Cut: Removing Vaccination
- Do-Cut Diff: What Changes?
- Soft Do: Attenuated Intervention
- Comparing Multiple Interventions
- Interpreting Counterfactuals
- Summary

### Introduction

One of the most powerful features of causal databases is the ability to reason about *interventions*. Pearl’s *do-calculus* formalizes the question: “What would happen if we *forced* variable X to a particular value?”

In CSQL.jl, interventions are implemented as SQL query rewrites that simulate cutting or attenuating outgoing edges from a concept. This vignette covers:

- **Hard do-cut** — completely remove a concept’s influence
- **Soft do** — attenuate (reduce) a concept’s influence
- **Do-cut diff** — compare baseline vs intervention

### Building the Atlas

We’ll use a classic epidemiological causal model:

```
using CSQL
```

```
builder = AtlasBuilder()
```

```
# Disease transmission chain
```

```

add_triple!(builder, "Pathogen", "increases", "Transmission";
             doc_id="doc1", score=0.95)
add_triple!(builder, "Contact rate", "increases", "Transmission";
             doc_id="doc1", score=0.87)
add_triple!(builder, "Transmission", "increases", "Infection prevalence";
             doc_id="doc1", score=0.92)
add_triple!(builder, "Infection prevalence", "increases", "Mortality";
             doc_id="doc2", score=0.78)

# Intervention: vaccination
add_triple!(builder, "Vaccination", "reduces", "Susceptibility";
             doc_id="doc2", score=0.88)
add_triple!(builder, "Vaccination", "increases", "Herd immunity";
             doc_id="doc2", score=0.91)
add_triple!(builder, "Susceptibility", "increases", "Transmission";
             doc_id="doc1", score=0.83)
add_triple!(builder, "Herd immunity", "reduces", "Transmission";
             doc_id="doc2", score=0.79)

# Healthcare
add_triple!(builder, "Healthcare", "reduces", "Mortality";
             doc_id="doc3", score=0.90)

csql = connect_csql()
build!(builder, csql.db)

SQLite.DB(":memory:")

```

## Baseline: The Causal Backbone

First, let's see the full causal backbone:

```

println("=== Baseline Backbone ===")
base = backbone(csql; limit=20)
for r in base
    println("  $(r.src) -[$(r.rel_type)]→ $(r.dst)  score=$(round(r.score_sum; digits=2))")
end
println("\nTotal edges: $(length(base))")

=== Baseline Backbone ===
pathogen -[INCREASES]→ transmission  score=0.95
transmission -[INCREASES]→ infection prevalence  score=0.92
vaccination -[INCREASES]→ herd immunity  score=0.91
healthcare -[REDUCES]→ mortality  score=0.9
vaccination -[REDUCES]→ susceptibility  score=0.88
contact rate -[INCREASES]→ transmission  score=0.87
susceptibility -[INCREASES]→ transmission  score=0.83

```

```

herd immunity -[REDUCES]→ transmission  score=0.79
infection prevalence -[INCREASES]→ mortality  score=0.78

```

Total edges: 9

## Hard Do-Cut: Removing Vaccination

The `do_cut` function removes all outgoing edges from a concept, simulating the question: “What would the causal landscape look like without vaccination?”

```

println("=== Do-Cut: Remove Vaccination ===")
cf = do_cut(csql, "vaccination"; limit=20)
for r in cf
    println("  $(r.src) -[$(r.rel_type)]→ $(r.dst)  score=$(round(r.score_sum; digits=2))")
end
println("\nEdges after intervention: $(length(cf))")

=== Do-Cut: Remove Vaccination ===
pathogen -[INCREASES]→ transmission  score=0.95
transmission -[INCREASES]→ infection prevalence  score=0.92
healthcare -[REDUCES]→ mortality  score=0.9
contact rate -[INCREASES]→ transmission  score=0.87
susceptibility -[INCREASES]→ transmission  score=0.83
herd immunity -[REDUCES]→ transmission  score=0.79
infection prevalence -[INCREASES]→ mortality  score=0.78

```

Edges after intervention: 7

Notice that vaccination’s effects (reducing susceptibility, increasing herd immunity) are gone — but the downstream concepts still exist via other causal pathways.

## Do-Cut Diff: What Changes?

`do_cut_diff` compares baseline and counterfactual, identifying which edges are lost:

```

diff = do_cut_diff(csql, "vaccination"; limit=20)
println("=== Edges Removed by do(¬vaccination) ===")
for r in diff.removed
    println("  $(r.src) -[$(r.rel_type)]→ $(r.dst)  score=$(round(r.score_sum; digits=2))")
end
println("\nBaseline: $(length(diff.baseline)) edges")
println("Counterfactual: $(length(diff.counterfactual)) edges")
println("Removed: $(length(diff.removed)) edges")

=== Edges Removed by do(¬vaccination) ===
vaccination -[INCREASES]→ herd immunity  score=0.91

```

```
vaccination -[REDUCES]→ susceptibility score=0.88
```

Baseline: 9 edges

Counterfactual: 7 edges

Removed: 2 edges

## Soft Do: Attenuated Intervention

Sometimes we don't want to completely remove a variable's influence — just reduce it. `soft_do` attenuates outgoing scores by a factor:

```
println("=== Soft Do: Reduce Pathogen influence by 80% ===")
soft = soft_do(csql, "pathogen"; attenuation=0.2, limit=20)
for r in soft
    score_str = round(r.score_sum_adj; digits=2)
    println("  $(r.src) -[$(r.rel_type)]→ $(r.dst) adj_score=$(score_str)")
end
```

```
=== Soft Do: Reduce Pathogen influence by 80% ===
transmission -[INCREASES]→ infection prevalence adj_score=0.92
vaccination -[INCREASES]→ herd immunity adj_score=0.91
healthcare -[REDUCES]→ mortality adj_score=0.9
vaccination -[REDUCES]→ susceptibility adj_score=0.88
contact rate -[INCREASES]→ transmission adj_score=0.87
susceptibility -[INCREASES]→ transmission adj_score=0.83
herd immunity -[REDUCES]→ transmission adj_score=0.79
infection prevalence -[INCREASES]→ mortality adj_score=0.78
pathogen -[INCREASES]→ transmission adj_score=0.19
```

Compare with baseline — pathogen's outgoing edge score is reduced to 20% of original, while all other edges retain full scores.

## Comparing Multiple Interventions

Let's compare the impact of different interventions:

```
println("=== Intervention Comparison ===\n")

interventions = ["vaccination", "healthcare", "pathogen"]
for concept in interventions
    diff = do_cut_diff(csql, concept; limit=20)
    println("do(~$(concept)):")
    println("  Edges removed: $(length(diff.removed))")
    if !isempty(diff.removed)
        total_lost = sum(r.score_sum for r in diff.removed)
        println("  Total score mass lost: $(round(total_lost; digits=2))")
    end
end
```

```

    println()
end

=== Intervention Comparison ===

do(~vaccination):
  Edges removed: 2
  Total score mass lost: 1.79

do(~healthcare):
  Edges removed: 1
  Total score mass lost: 0.9

do(~pathogen):
  Edges removed: 1
  Total score mass lost: 0.95

```

This reveals which concepts have the most causal influence: removing them disrupts the most edges and score mass.

## Interpreting Counterfactuals

The do-cut analysis answers practical questions:

| Question                                        | CSQL Operation                                              |
|-------------------------------------------------|-------------------------------------------------------------|
| “What if there were no vaccination?”            | <code>do_cut(csql, "vaccination")</code>                    |
| “What if the pathogen were less transmissible?” | <code>soft_do(csql, "pathogen";<br/>attenuation=0.3)</code> |
| “Which intervention matters most?”              | Compare <code>do_cut_diff</code> across interventions       |
| “What causal paths survive without X?”          | <code>do_cut + causal_paths</code> on result                |

## Summary

This vignette demonstrated:

- **do\_cut** — hard intervention removing all outgoing edges
- **soft\_do** — partial attenuation of outgoing influence
- **do\_cut\_diff** — comparison of baseline vs intervention
- **Multi-intervention comparison** — ranking concepts by causal importance

Next: Provenance and LCMs covers tracking the source documents behind causal claims.